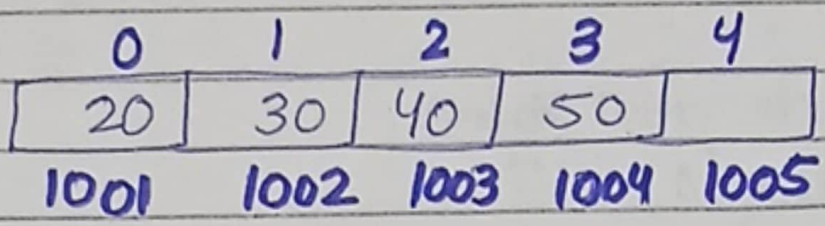
Task 1 :

Insert is at the beginning of the array:

Intial array: 20, 30, 40, 50.

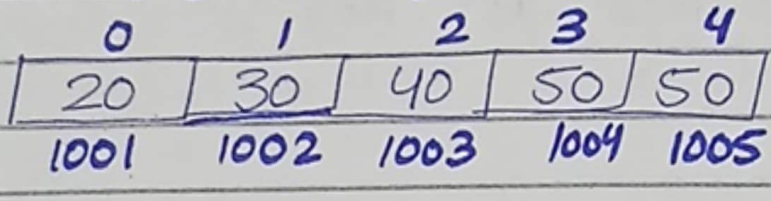
Size(n) = 4

Memory Address : 1001 1002 1003 1004

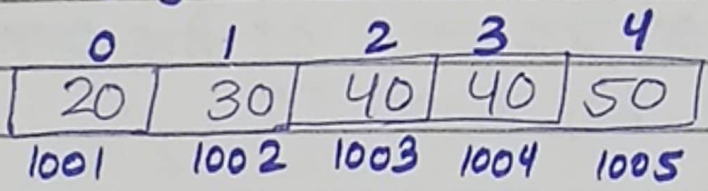


Shift elements to right.

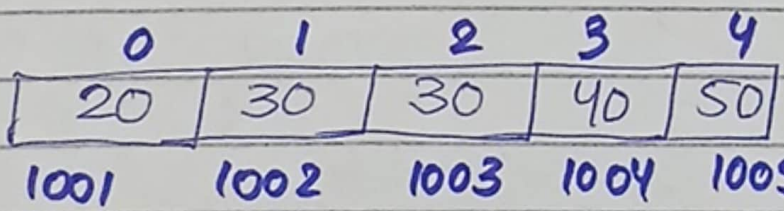
→ arr[4] = arr[3]



→ arr[3] = arr[2]

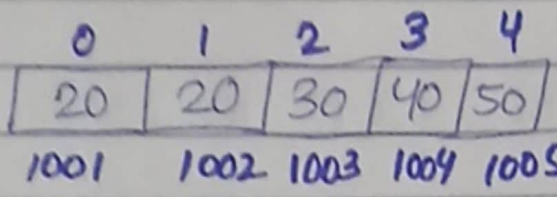


→ arr[2] = arr[1]

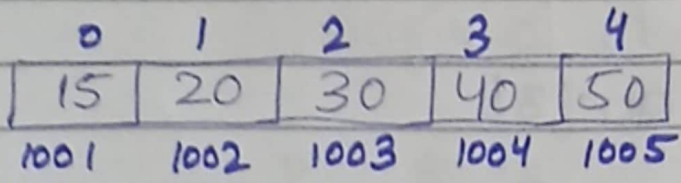


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→ arr[1] = arr[0]



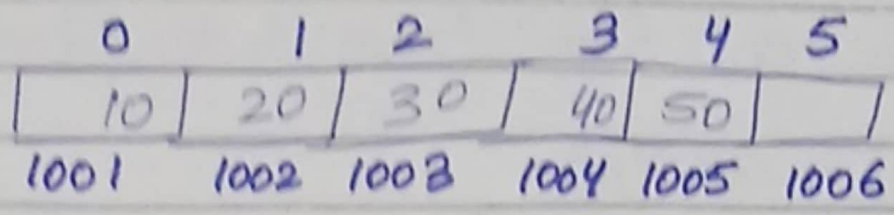
→ Insert new value:
arr[0] = 15.



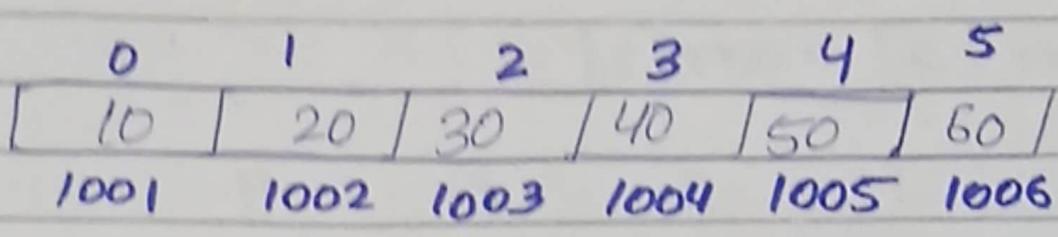
Result : 15 → 20 → 30 → 40 → 50

Task 2:
Insert 60 at the end of the array:
10, 20, 30, 40, 50

Intial Array:
Size(n) = 5
Memory Address: 1001 1002 1003
1004 1005



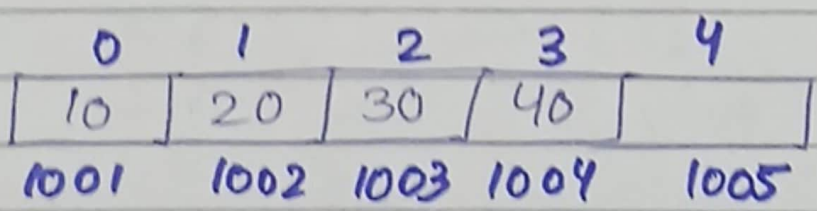
→ Direct Insertion:
arr[5]=60



Result: 10 → 20 → 30 → 40 → 50 → 60.

Task 3:
Insert 25 at index 2 in:
10, 20, 30, 40.

Intial Array:
Size(n) = 4
target index = 2.



Shift Element Right from Index 2.

→ arr[4] = arr[3]

0	1	2	3	4
10	20	30	40	40
1001	1002	1003	1004	1005

→ arr[3] = arr[2]

0	1	2	3	4
10	20	30	30	40
1001	1002	1003	1004	1005

→ arr[2] = 25

0	1	2	3	4
10	20	25	30	40
1001	1002	1003	1004	1005

Result : 10 → 20 → 25 → 30 → 40.

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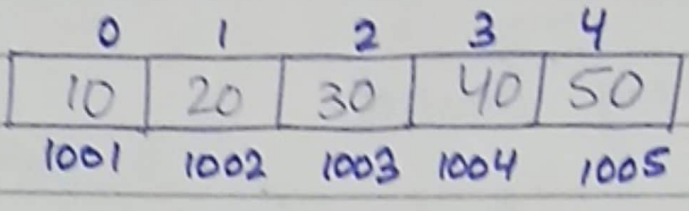
Task 4:

Delete the first element from:

10, 20, 30, 40, 50.

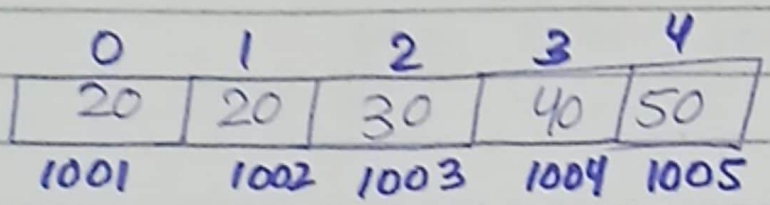
Initial Array:

Size (n) = 5

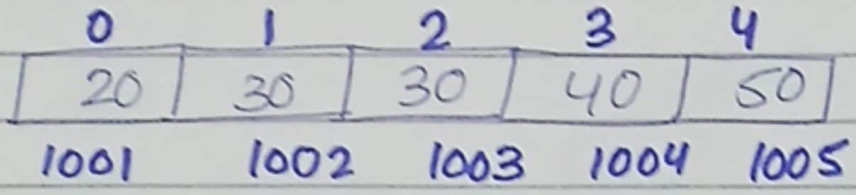


→ Shift elements left:

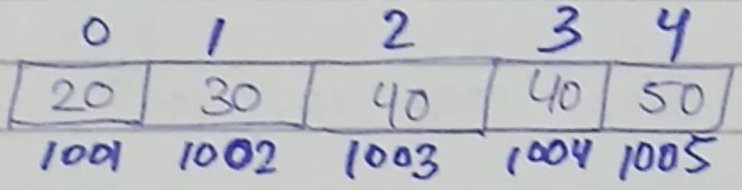
arr[0] = arr[1]



→ arr[1] = arr[2]

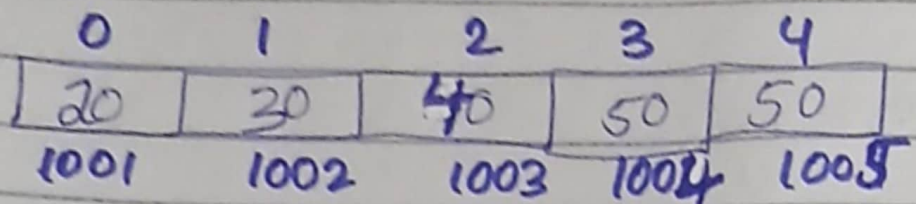


→ arr[2] = arr[3]



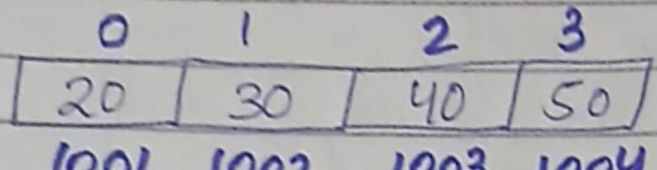
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→ arr[3] = arr[4]



Update Array Baund:

→ n = n - 1 (Active size = 4)



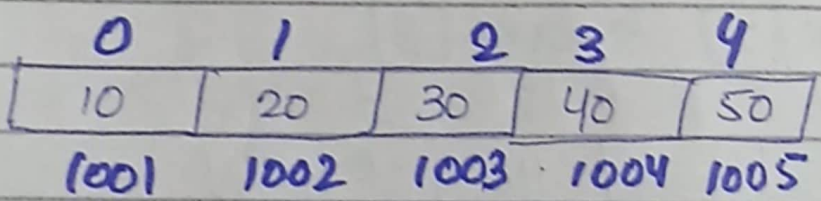
Result : 20 → 30 → 40 → 50.

Task 5:

Delete the last element from 10, 20, 30, 40, 50.

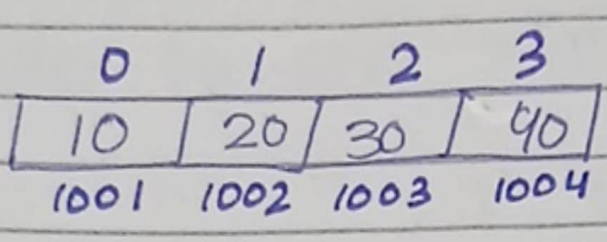
Intial Array:

Size (n) = 5



Update size pointer:

→ $n = n - 1$ (Active size = 4)



Result : 10 → 20 → 30 → 40.

Task 6 :

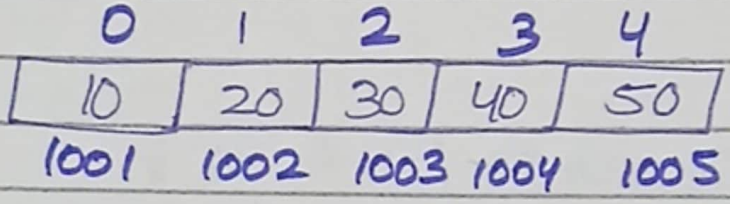
Delete the element at index 2 from:

10, 20, 30, 40, 50.

Initial Array :

Size (n) = 5

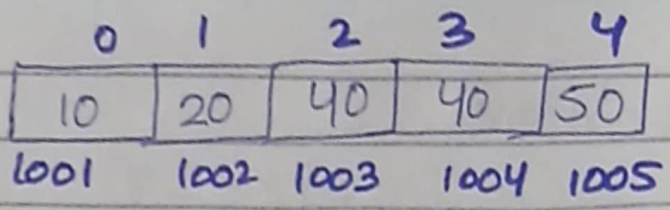
target index = 2 (Value 30)



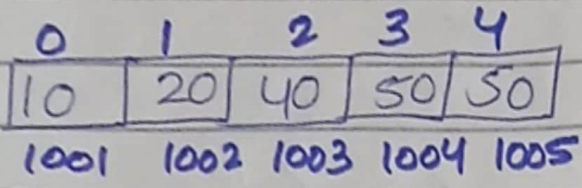
Shift elements left from Index 2:

→ arr[2] = arr[3]

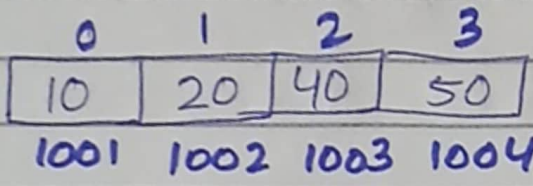
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→ arr[3] = arr[4]



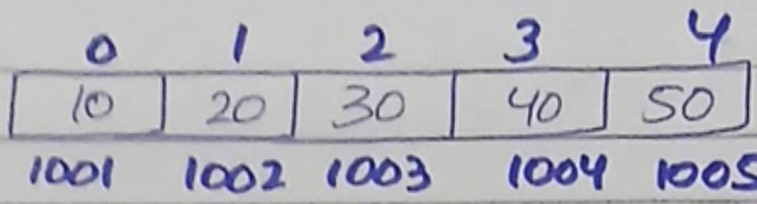
Update Array Bounds:
→ n = n-1 (Active size = 4)



Result: 10 → 20 → 40 → 50.

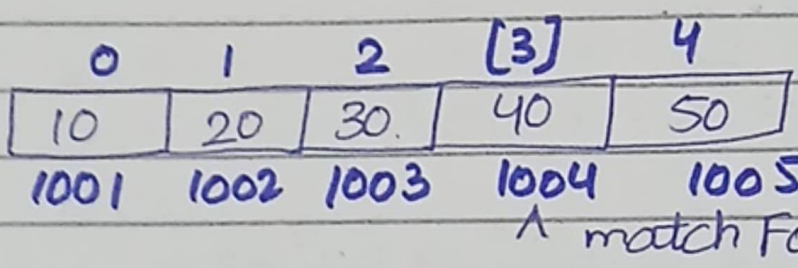
Task 7:
Linear Search to Find 40 in:
10, 20, 30, 40, 50

Target Value = 40
Intial Array:
Size (n) = 5



→ Step by step Traversal:

- i = 0
  arr [0] = 10 → False (move to next)
- i = 1
  arr [1] = 20 → False (move to next)
- i = 2
  arr [2] = 30 → False (move to next)
- i = 3
  arr [3] = 40 → True (Match Found)

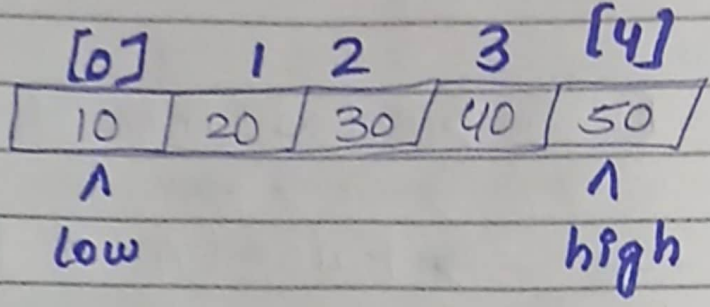


Result: Value 40 found at index 3.

Task 8:
Binary search to find 40 in sorted array: 10, 20, 30, 40, 50.

Target Value = 40.

Intial search Bounds:
low = 0 , high = 4.



Iteration 1:

$$mid = \frac{low + high}{2}$$

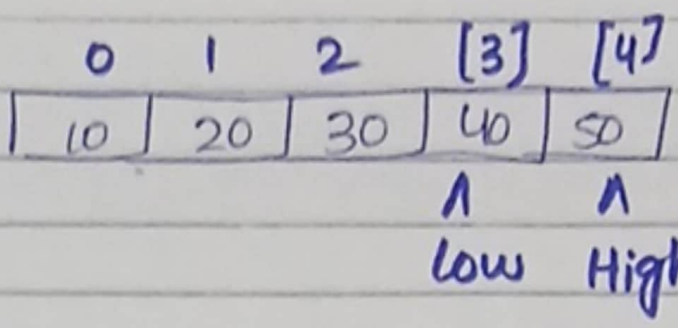
$$= \frac{0+4}{2} = 2$$

arr[mid] ⇒ arr[2] = 30

Check 30 == 40 → False.

Check 30 < 40 → False

low = mid + 1 ⇒ 2 + 1
= 3



Iteration 2:

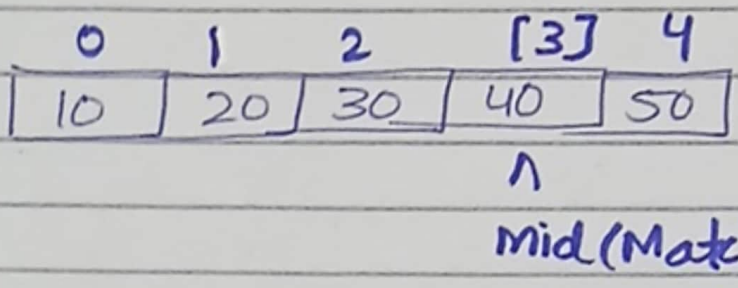
$$mid = \frac{low + high}{2}$$

$$= \frac{3+4}{2}$$

$$= \frac{7}{2} \approx 3$$

arr[mid] => a[3] = 40.

check 40 == 40 -> True (Match Found)



Result: Value 40 fount at index 3.